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The effect of pedometer use in combination with cognitive and behavioral support materials to promote physical activity☆

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Abstract

Objective: 1. To demonstrate that combining pedometer use with cognitive and behavioral support materials has a positive effect on physical activity (PA) and attitudes towards pedometer use. 2. To investigate how familiar the study sample is with pedometers and the '10,000 steps/day' recommendation.

Methods: From a random sample, drawn from the phone book, 304 volunteered (18–75 year) to complete a questionnaire about familiarity with pedometers and the '10,000 steps/day' recommendation. A sample of 103 participants agreed to wear a pedometer for 3 weeks, and was randomly assigned to a condition with cognitive and behavioral support materials ($n = 51$) or without these materials ($n = 52$). Participants completed the International Physical Activity Questionnaire before and after 21 days of pedometer use and an additional questionnaire on the attitudes towards pedometer use.

Results: More than 58% had never heard of a pedometer. In both conditions, walking ($F = 10$, $p = 0.002$), moderate PA ($F = 11$, $p = 0.001$), and vigorous PA ($F = 14$, $p < 0.001$) increased over time, however no interaction effects could be found. Significantly more participants in the condition with support materials had a positive attitude towards pedometer use.

Conclusion: Wearing a pedometer, with or without support materials, may increase PA. In our study, cognitive and behavioral support materials only affected attitudes towards pedometer use.

Practice implications: More research is needed to investigate the effect of combining pedometer use with support materials on a longer time base and in less motivated people.

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Keywords: Step counter; Self-monitoring; '10,000 steps/day'

1. Introduction

International health guidelines recommend the accumulation of minimum 30 min of at least moderate-intensity PA on most, preferably all, days of the week [1–3]. With the increased use of pedometers as objective PA measurement, step count goals have been used [4,5], and '10,000 steps/day' has become a popular PA target [6,7].

Using a pedometer can provide a reminder to be more active, however the question is whether simply providing a pedometer is enough to increase PA. There is also a need to increase knowledge regarding health benefits of PA, to motivate individuals [8], and to reinforce commitment to PA routine. According to Robinson and Rogers, and Dunn et al., cognitive and behavioral strategies (self-monitoring, goal-setting techniques) are essential [8,9]. Furthermore, Tudor-Locke and Bassett believe that providing guidelines and programs are also valuable [10]. In previous interventions, pedometers were used in combination with these strategies [11–16]. A different approach is the simple distribution of pedometers with limited information. The main purpose was to evaluate if combining pedometer use with cognitive and behavioral support materials, such as information on health, PA, and the '10,000 steps/day' recommendation, goal-setting, and self-monitoring in log books, has a positive effect on PA levels and on the attitudes towards pedometer use.

The use of pedometers and step count goals is popular in America, Canada, and Australia [17–19], but still novel and only recently (2–3 years) gaining popularity in Belgium,

☆ I confirm all patient/personal identifiers have been removed or disguised so the patient/person(s) described are not identifiable and cannot be identified through the details of the story.

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Europe. Consequently, an additional purpose was to evaluate how familiar the study population is with pedometers and the '10,000 steps/day' recommendation.

2. Methods

2.1. Procedures and participants

From the phone book covering the west part of East-Flanders, 724 telephone numbers of citizens of Zottegem, Belgium were selected by simple randomization (selection of the first name and every tenth name below the previous one). Within two attempts, 447 individuals older than 18 could be reached. A sample of 304 volunteered to complete a telephone interview relating to familiarity with pedometers and the '10,000 steps/day' recommendation (see Table 1). Participants prepared to use a pedometer for three consecutive weeks ($n = 103$) were asked to complete the International Physical Activity Questionnaire (IPAQ) at the end of the interview and after 3 weeks of pedometer use. Two-hundred-and-one participants were not willing to use a pedometer (66% ''no interest in pedometer use'', 24% ''no time to wear a pedometer'' and 16% ''another reason'').

After the interview, participants who agreed to wear a pedometer were randomly assigned to a condition using a pedometer with cognitive and behavioral support materials ('condition +') ($n = 51$) or a condition without those materials ('condition −') ($n = 52$) (1st participant: 'condition −', 2nd: 'condition +', 3rd: 'condition −', ...). All participants were sent a package including: a pedometer, an informed consent form, a protocol describing how and when to use the pedometer, a questionnaire about the attitudes towards pedometer use, the IPAQ (both to complete after 3 weeks of pedometer use), and a preaddressed and stamped envelope to return the pedometer and questionnaires.

The 'condition +' was sent a brochure (designed by the researchers, based on items from the social cognitive theory [20]) with information on the importance of PA for health (two pages), the '10,000 steps/day' recommendation (three pages), and 20 tips on how to increase PA (e.g. 'take the stairs'). On the last three pages, participants were asked to set goals and to complete a daily step log book (to be returned to the researchers). The first week served to determine the baseline step level and in the second and third week participants were asked to increase their baseline step level by 2000 steps/day if the level was below 10,000 steps/day. If it was above the recommended 10,000 steps/day, participants were encouraged to maintain their physically active lifestyle.

2.2. Instruments

2.2.1. Telephone interview

First, gender, age group, educational level (elementary or high school = 'low-educated'; college or university = 'high-educated'), and employment status were asked. Then, individuals were asked close-ended questions (yes/no) related to knowledge about, and familiarity with pedometers and the steps/day-guideline (see Table 2).

2.2.2. IPAQ

The IPAQ short-form telephone version (before pedometer use) and self-administered version (after pedometer use) [http://www.ipaq.ki.se] were used to assess PA in the last 7 days. Both versions give similar results [21] and the IPAQ has been shown to be a valid and reliable instrument for measuring PA [21,22].

2.2.3. Pedometers

The Yamax Digiwalker SW-200 (Yamax, Tokyo, Japan) was used in the present study as it is known to be one of the best

Table 1
Descriptive characteristics

Variables	**Total**	Not willing to wear pedometer	Agreed to wear pedometer	χ^2 (p)	Condition +	Condition −	χ^2 (p)
	***n* (%)**	*n* (%)	*n* (%)		*n* (%)	*n* (%)	
n	**304 (100)**	201/304 (66.1)	103/304 (33.9)		51/103 (49.5)	52/103 (50.5)	
Gender				4.9 (*)			0.5 (ns)
Male	**136/304 (44.7)**	99/136 (72.8)	37/136 (27.2)		20/51 (39.2)	17/52 (32.7)	
Female	**168/304 (55.3)**	102/168 (60.7)	66/168 (39.3)		31/51 (60.8)	35/52 (67.3)	
Age				3.5 (ns)			0.8 (ns)
18–25 years	**79/304 (26.2)**	54/79 (68.4)	25/79 (31.6)		12/51 (23.5)	13/52 (25)	
26–55 years	**164/304 (54.3)**	101/164 (61.6)	63/164 (38.4)		30/51 (58.8)	33/52 (63.5)	
56–75 years	**59/304 (19.5)**	44/59 (74.6)	15/59 (25.4)		9/51 (17.6)	6/52 (11.5)	
Level of education				0.3 (ns)			0.8 (ns)
Low	**192/304 (63.2)**	129/192 (67.2)	63/192 (32.8)		29/51 (56.9)	34/52 (65.4)	
High	**112/304 (36.8)**	72/112 (64.3)	40/112 (35.7)		22/51 (43.1)	18/52 (34.6)	
Occupation				3.6 (ns)			0.6 (ns)
Employed	**213/304 (79.4)**	135/213 (63.4)	78/213 (36.6)		40/51 (78.4)	38/52 (73.1)	
Unemployed	**51/304 (16.1)**	34/51 (66.7)	17/51 (33.3)		7/51 (13.7)	10/52 (19.2)	
Retired	**40/304 (12.5)**	32/40 (80)	8/40 (20)		4/51 (7.8)	4/52 (7.7)	

ns, not significant ($p > 0.05$); *$p \leq 0.05$.

Table 2
Familiarity with PM and '10,000 steps/day' recommendation

	Total *n* (% yes)	Not willing to wear PM *n* (% yes)	Agreed to wear PM *n* (% yes)	χ^2 (*p*)
Have you already heard of a step counter or PM?	**127/304 (41.8)**	72/201 (35.8)	55/103 (53.4)	8.7 (**)
Where?				
Television	**31/127 (24.4)**	20/72 (27.8)	11/55 (20)	
Radio	**4/127 (3.1)**	3/72 (4.2)	1/55 (1.8)	
Magazine	**25/127 (19.7)**	14/72 (19.4)	11/55 (20)	
Home	**15/127 (11.8)**	10/72 (13.9)	5/55 (9.1)	
Work	**11/127 (8.7)**	6/72 (8.3)	5/55 (9.1)	
Family	**31/127 (24.4)**	16/72 (22.2)	15/55 (27.3)	
Other	**10/127 (7.9)**	3/72 (4.2)	7/55 (12.7)	
Have you already seen a PM?	**66/304 (21.7)**	36/201 (17.9)	30/103 (29.1)	5.0 (*)
Where?				
Television	**16/66 (24.2)**	8/36 (22.2)	8/30 (26.7)	
Home	**13/66 (19.7)**	9/39 (25)	4/30 (13.3)	
Work	**8/66 (12.1)**	3/36 (8.3)	5/30 (16.7)	
Family	**20/66 (30.3)**	11/36 (30.6)	9/30 (30)	
Other	**9/66 (13.6)**	5/36 (13.8)	4/30 (13.3)	
Do you know the function of a PM?	**114/304 (37.5)**	63/201 (31.3)	51/103 (49.5)	9.6 (**)
Where from?				
Television	**24/114 (21.1)**	12/63 (19)	12/51 (23.5)	
Radio	**2/114 (1.8)**	2/63 (3.2)	0 (0)	
Magazine	**20/114 (17.5)**	11/63 (17.5)	9/51 (17.6)	
Home	**17/114 (14.9)**	12/63 (19)	5/51 (9.8)	
Work	**13/114 (11.4)**	7/63 (11.1)	6/51 (11.8)	
Family	**27/114 (23.7)**	14/63 (22.2)	13/51 (25.5)	
Other	**11/114 (9.6)**	5/63 (7.9)	6/51 (11.8)	
Do you know how to use a PM properly?	**42/304 (13.8)**	21/201 (10.4)	21/103 (20.4)	5.7 (*)
Where from?				
Television	**5/42 (11.9)**	2/21 (9.5)	3/21 (14.3)	
Magazine	**5/42 (11.9)**	3/21 (14.3)	2/21 (9.5)	
Home	**10/42 (23.8)**	7/21 (33.3)	3/21 (14.3)	
Work	**8/42 (19)**	3/21 (14.3)	5/21 (23.8)	
Family	**13/42 (31)**	5/21 (23.8)	8/21 (38.1)	
Other	**1/42 (2.4)**	1/21 (4.8)	0 (0)	
Do you know how many steps/day are health beneficial?	**53/304 (17.4)**	28/201 (13.9)	25/103 (24.3)	5.1 (*)
Idea of this number (mean ± S.D.)	**7707 ± 3591**	7221 ± 3545	8250 ± 3636	$t = 1.0$ (ns)

PM, pedometer; ns = not significant ($p > 0.05$); *$p \leq 0.05$; **$p \leq 0.01$.

pedometers in regard to its accuracy and reliability for counting steps [23,24].

2.2.4. Questionnaire on attitudes towards pedometer use

The questionnaire included close-ended questions relating to pedometer use. Participants in the 'condition +' were asked to complete four additional items (see Table 3).

2.3. Data analysis

All data were analyzed using SPSS 12.0 for Windows. Chi-square tests were used to analyze descriptive characteristics, familiarity with pedometers, and attitudes. A repeated measures MANOVA was used to evaluate the effect over time, the differences between the two conditions, and the interaction effect (time × condition) of the truncated (maximum 2 h/day) and log transformed IPAQ values. For reasons of clarity and comparability, the numbers in Section 3 and in Table 3 are the means and standard deviations (S.D.) of the non-transformed IPAQ data. Independent samples *t*-tests were used to compare the reported number of steps/day between different groups. The difference in step counts after 3 weeks of pedometer use in participants in the 'condition +' was evaluated with a paired-sample *t*-test. The alpha level was set at 0.05, except for multiple tests ($\alpha = 0.01$).

3. Results

3.1. Familiarity with pedometers and the '10,000 steps/day' recommendation (see Table 2)

Significantly more individuals who agreed to wear a pedometer, than those not willing to wear one, had already heard of a pedometer, had already seen a pedometer, knew the function of the device, knew how to use it properly, and reported knowing how many steps/day are health beneficial.

Table 3
Attitudes towards pedometer use

	Condition + $n = 51$ n (%)	Condition − $n = 52$ n (%)	χ^2 (p)
I wore the pedometer everyday during the 3 weeks			1.4 (ns)
"Totally not agree" and "not agree"	5/50 (10)	9/49 (18.4)	
"Agree" and "fully agree"	45/50 (90)	40/49 (81.6)	
I wore the pedometer only the first couple of days			4.1 (*)
"Totally not agree" and "not agree"	49/50 (98)	42/48 (87.5)	
"Agree" and "fully agree"	1/50 (2)	6/48 (12.5)	
I found it pleasant to wear the pedometer			7.5 (**)
"Totally not agree" and "not agree"	1/50 (2)	9/48 (18.8)	
"Agree" and "fully agree"	49/50 (98)	39/48 (81.3)	
The pedometer bothered me during the day			2.9 (ns)
"Totally not agree" and "not agree"	48/51 (94.1)	40/48 (83.3)	
"Agree" and "fully agree"	3/51 (5.9)	8/48 (16.7)	
During the day, I was influenced by the number of steps			3.4 (ns)
"Totally not agree" and "not agree"	33/51 (64.7)	39/48 (81.3)	
"Agree" and "fully agree"	18/51 (35,3)	9/48 (18.8)	
I walked more because of the pedometer			2.9 (ns)
"Totally not agree" and "not agree"	37/50 (74)	42/48 (87.5)	
"Agree" and "fully agree"	13/50 (26)	6/48 (12.5)	
I would be prepared to use the pedometer for a longer period			5.6 (*)
"Totally not agree" and "not agree"	9/50 (18)	19/48 (39.6)	
"Agree" and "fully agree"	41/50 (82)	29/48 (60.4)	
I would buy a pedometer			4.2 (*)
"Totally not agree" and "not agree"	22 (45.8)	32 (66.7)	
"Agree" and "fully agree"	26 (54.2)	16 (33.3)	
I found the information on pedometer use clear			1.2 (ns)
"Totally not agree" and "not agree"	5/48 (9.8)	2/48 (4.2)	
"Agree" and "fully agree"	46/48 (90.2)	46/48 (95.8)	
I still have questions about pedometer use			17.8 (***)
"Totally not agree" and "not agree"	46/51 (90.2)	22/43 (51.2)	
"Agree" and "fully agree"	5/51 (9.8)	21/43 (48.8)	
I believe to know how many steps/day are health beneficial	35/49 (71.4)	22/51 (43.1)	8.1 (**)
Idea of number of steps (mean ± S.D.)	9714 ± 1274	9091 ± 2667	$t = 1.0$ (ns)
I find it a good idea to increase my PA by means of a pedometer and the '10,000 steps/day' recommendation.			
"Totally not agree" and "not agree"	5/51 (9.8)		
"Agree" and "fully agree"	46/51 (90.2)		
I find the combination of information on PA and health, pedometer use, and self-monitoring methods motivating to increase my PA			
"Totally not agree" and "not agree"	8/51 (15.7)		
"Agree" and "fully agree"	43/51(84.4)		
I find the combination of information on PA and health, pedometer use, and self-monitoring methods not motivating because I could not reach the target of '10,000 steps/day'.			
"Totally not agree" and "not agree"	45/51 (78.4)		
"Agree" and "fully agree"	6/51(11.8)		
I find it annoying to register the steps at the end of the day.			
"Totally not agree" and "not agree"	49/51 (96.1)		
"Agree" and "fully agree"	2/51 (3.9)		

ns, not significant ($p > 0.05$); *$p \leq 0.05$; **$p \leq 0.01$; ***$p \leq 0.001$.

3.2. Amount of PA

Multivariate analyses showed no significant interaction effect ($F = 0.8$, $p > 0.05$), nor an effect for the different conditions ($F = 0.4$, $p > 0.05$). However, a strong time effect ($F = 8.3$, $p < 0.001$) was found, showing increases in minutes/week walking, minutes/week moderate, and minutes/week vigorous PA in both conditions (Table 4).

3.3. Attitudes towards pedometer use (see Table 3)

More participants in the 'condition −' than in the 'condition +' reported that they wore the pedometer only the first couple of days, they found it not pleasant to wear the pedometer, and they still had questions about pedometer use. Significantly more participants in the 'condition +' reported to be prepared to wear the pedometer for a longer period, would buy a pedometer, and

Table 4
Amount of physical activity

Min/week	Before pedometer use					After pedometer use					F (p)[a]
	Total	Condition +	n	Condition −	n	**Total**	Condition +	n	Condition −	n	Time effect
Sitting (mean min/day ± S.D.)	**370 ± 186**	396 ± 218	(41)	371 ± 148	(39)	**377 ± 176**	363 ± 190	(41)	393 ± 161	(39)	0.2 (ns)
Walking (mean ± S.D.)	**212 ± 290**	211 ± 290	(40)	212 ± 294	(42)	**267 ± 305**	257 ± 287	(40)	276 ± 324	(42)	6.1 (*)
Moderate PA (mean ± S.D.)	**56 ± 137**	58 ± 151	(50)	54 ± 121	(45)	**129 ± 215**	105 ± 189	(50)	157 ± 240	(45)	12.9 (***)
Vigorous PA (mean ± S.D.)	**43 ± 95**	39 ± 87	(41)	46 ± 102	(39)	**73 ± 130**	52 ± 115	(41)	94 ± 143	(39)	6.5 (*)

ns, not significant ($p > 0.05$); *$p \leq 0.05$; ***$p \leq 0.001$.
[a] F and p values represent difference over time (before vs. after) for both conditions (total).

believed to know how many steps are health beneficial. There was a trend that more participants in the 'condition +' were not bothered by the pedometer ($p = 0.088$), were influenced by the number of steps ($p = 0.065$), and reported to have walked more because of wearing a pedometer ($p = 0.091$).

3.4. Perception of the cognitive and behavioral support materials in the 'condition +' (see Table 3)

Results showed that 90% found it a good idea to increase PA by means of a pedometer and the '10,000 steps/day' recommendation. Forty-three found that the whole concept worked to motivate an increase in daily PA. About 12% found it not motivating because '10,000 steps/day' could not be reached. Two participants (3.9%) found it annoying to register the steps taken at the end of the day. About 94% ($n = 48$) completed the daily step log. There was an increase in average step counts (first week: 9291 ± 3526; third week: 10,010 ± 3250) ($t = 1.7$, $p = 0.087$).

4. Discussion and conclusion

4.1. Discussion

Despite the growing popularity of pedometers and the '10,000 steps/day' recommendation, participants were not entirely familiar with the concept. Furthermore, it was hypothesized that providing cognitive and behavioral support materials would have a positive effect on the amount of PA. However, the 'condition +' did not report more PA over time than the 'condition −'. Although previous pedometer interventions [11–16] used pedometers in combination with cognitive and behavioral strategies, present results showed that simply providing a valid pedometer without support materials to persons willing to use it, can result in a PA increase. Note that besides walking, also moderate and vigorous PA increased, suggesting that pedometer use can generally sensitize PA engagement. In contrast to the present results, Glazener et al. [25] suggested that the use of a pedometer alone without specific guidelines or individual step goals, may not be as effective in increasing PA as when recommendations or targets are being used.

While providing support materials had no additional effect on PA, it had a positive influence on the attitudes towards pedometer use. More than 84% of the participants in the 'condition +', believed that the whole concept motivated them to increase their PA. There was a trend that more participants in the 'condition +' reported to be influenced by the number of steps, and to walk more because of the pedometer. However, the majority in both conditions reported not being influenced by their step counts and not walking more, yet their PA increased. Possibly pedometer use not only sensitized for steps or walking, but also for general PA engagement.

Overall, present results should be interpreted with caution. Findings are limited in generalizability. The impact of support materials can be different in less motivated people or in persons not contacted or invited in advance and for example just given a pedometer with no further information. Secondly, the duration of the present study was only three weeks. Extra research is necessary to examine the use of promotional materials on a longer time base. Furthermore, there is no information on the actual use of the support materials. Further studies can evaluate the adherence to support materials. Finally, social desirability may have influenced the findings of the present study.

4.2. Conclusion

Wearing a valid pedometer (with or without support materials) may increase PA on a short time base. Furthermore, it was found that in our study support materials only affected attitudes towards pedometer use.

4.3. Practical implications

The present findings can be relevant to the merchandising industry, other researchers and practitioners who develop lifestyle PA programs, interventions, and promotion targets.

References

[1] US Department of Health and Human Services. Physical activity and health: a report of the surgeon general. Atlanta, GA US Department of Health and Human Services, Centers for Disease Control and Prevention,

National Center for Chronic Disease Prevention and Health Promotion. J Am Med Assoc 1996;276:522.
[2] Pate RR, Pratt MS, Blair SN, Haskell WL, Macera CA, Bouchard C, Buchner D, Ettinger W, Heath GW, King AC, Kriska A, Leon AS, Marcus BH, Morris J, Paffenbarger RS, Patrick K, Pollock ML, Rippe JM, Sallis J, Wilmore JH. Physical activity and public health: a recommendation from the Centers for Disease Control and Prevention and the American College of Sports Medicine. J Am Med Assoc 1995;273:402–7.
[3] American College of Sports Medicine. ACSM's guideline for exercise testing and prescription, 6th ed. Philadelphia: Lippincott Williams and Wilkins; 2000. p. 5–7.
[4] Swartz AM, Strath SJ, Bassett DR, Moore JB, Redwine BA, Groer M, Thompson DL. Increasing daily walking improves glucose tolerance in overweight women. Prev Med 2003;37:356–62.
[5] Jensen GL, Roy M, Buchanan AE, Berg MB. Weight loss intervention for obese older women: improvements in performance and function. Obes Res 2004;12:1814–20.
[6] Hatano Y. Use of the pedometer for promoting daily walking exercise. ICHPER 1993;29:4–8.
[7] Wilde BE, Sidman CL, Corbin CB. A 10,000-step count as a physical activity target for sedentary women. Res Q Exerc Sport 2001;72: 411–4.
[8] Robinson JI, Rogers MA. Adherence to exercise programmes: recommendations. Sports Med 1994;17:39–52.
[9] Dunn AL, Anderson RE, Jakicic JM. Lifestyle physical activity interventions: history, short- and long-term effects, and recommendations. Am J Prev Med 1998;15:398–412.
[10] Tudor-Locke CE, Bassett DR. How many steps/day are enough? Preliminary pedometer indices for public health. Sports Med 2004;34: 1–8.
[11] Chan CB, Ryan DA, Tudor-Locke CE. Health benefits of a pedometer-based physical activity intervention in sedentary workers. Prev Med 2004;39:1215–22.
[12] Miller R, Brown WJ. Steps and sitting in a working population. Int J Behav Med 2004;11:219–24.
[13] Tudor-Locke CE, Myers AM, Rodger NW. Development of a theory-base daily activity intervention for individuals with type 2 diabetes. Diabetes Educ 2001;27:85–93.
[14] Schofield L, Mummery WK, Schofield G. Effects of a controlled pedometer-intervention trial for low-active adolescent girls. Med Sci Sports Exerc 2005;37:1414–20.
[15] Brown W, Mummery K, Eakin E, Schofield G. 10,000 steps Rockhampton: evaluation of a whole community approach to improving population levels of physical activity. J Phys Act Health 2006;3.
[16] Dinger M, Heesch K, Cipriani G, Qualls M. Comparison of two email-delivered, pedometer-based interventions to promote walking among insufficiently active women. J Sci Med Sport 2007;10:297–302.
[17] Kosta E. Make every step count. Walking 2001;16:54–61.
[18] Krucoff C. Popular, low-cost pedometers: 10,000 steps to a better health. The Seattle Times 5 December, 1999.
[19] Dietz WH. Canada on the move: a novel effort to increase physical activity among Canadians. Can J Public Health 2006;97:S3–4.
[20] Bandura A. Social cognitive theory: an agentive perspective. Annu Rev Psychol 2001;52:1–26.
[21] Craig CL, Marshall AL, Sjöström M, Bauman AE, Booth ML, Ainsworth BE, Pratt M, Ekelund U, Yngve A, Sallis JF, Oja P. International Physical Activity Questionnaire: 12-country reliability and validity. Med Sci Sports Exerc 2003;35:1381–95.
[22] Vandelanotte C, De Bourdeaudhuij I, Sallis J, Philippaerts R, Sjöström M. Reliability and validity of a computerised and Dutch version of the International Physical Activity Questionnaire (IPAQ). J Phys Act Health 2005;2:63–75.
[23] Schneider PL, Crouter SE, Lukajic O, Bassett DR. Accuracy and reliability of 10 pedometers for measuring steps over a 400 m walk. Med Sci Sports Exerc 2003;35:1779–84.
[24] Crouter SE, Schneider PL, Karabulut M, Bassett DR. Validity of 10 electronic pedometers for measuring steps, distance and energy cost. Med Sci Sports Exerc 2003;35:1455–60.
[25] Glazener H, DeVoe D, Nelson T, Gotshall R. Changes in physical activity influenced by using a pedometer. J Hum Mov Stud 2004;46:473–82.